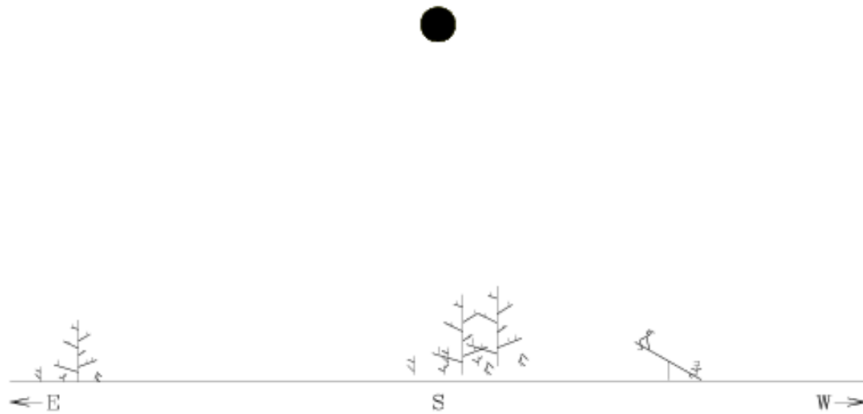


HW 1.4a

Name(s): _____

1. Is the Moon always visible at night?
2. If the Moon is in the third quarter phase today, will it be more or less illuminated tomorrow?
3. If the new Moon rose nine hours ago, what time of the day or night is it?
4. This figure shows the location and phase of the Moon as we look from our position in the northern hemisphere toward the southern sky, with the horizon stretching from east (on the left) to west (on the right). What time of the night or day is shown?



5. If the first quarter Moon will set in three hours, what time of the day or night is it?

6. Approximately how long does it take for the moon to change between two successive phases.
7. If the Moon lies visible above the horizon in the sky from noon (12pm) till midnight (12am), what phase will it have?
8. Does the Moon ever **transit** simultaneously for observers located in New York and in California? *An object that is transiting is at its highest location in the sky that night.*
9. If you observe the Moon in the figure below setting on the western horizon, what time of the day or night is it?



10. If the Moon is located as shown in the figure below, where would it be located in the sky at 6 pm?

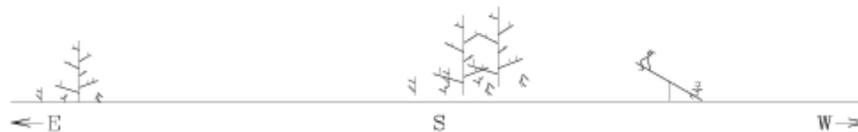


11. If you observe the Moon in a certain phase one night, will it show the same phase to a friend on the other side of the world who observes it twelve hours later?

12. If the Moon is located as shown in the figure below, will it be more or less illuminated tomorrow?

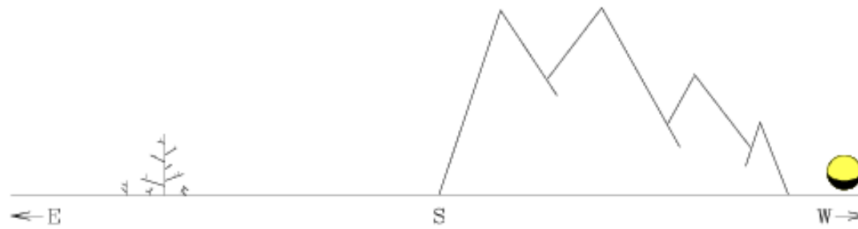


13. This figure shows the location of the Moon in the sky as we look from our position in the northern hemisphere toward the southern sky, with the horizon stretching from east (on the left) to west (on the right). If the figure shows the location of the Moon at noon (3pm), what phase must it exhibit?



14. Sarah on the Earth sees a waxing crescent moon. What phase Earth would an Astronaut on the Moon see?

15. This figure shows the location and phase of the Moon as we look from our position in the northern hemisphere toward the southern sky, with the horizon stretching from east (on the left) to west (on the right). What time of the night or day is shown?



16. If the Moon is moving down through the sky (no longer overhead, well above the western horizon) in the sky in the morning (9am), what phase will it have?
17. How long, in days, does it take the Moon to shift from a third quarter phase to the next Waxing Gibbous phase?
18. The moon is always _____% illuminated
19. Is the illuminated side of the Moon always the same as the visible side?
20. In one sentence, why does the moon have phases?